TOOLKIT

Stage a Speed Interviewing Marathon

—Jenna Carpenter and Michael Dorff

Mathematics majors interviewing for that first "real" job often are not aware of the array of options available to them or even the basics of job searching. Although some campus career centers may offer seminars on "how to interview," they are often focused on large corporations for the campus career fair. These seminars generally do not include information on how to find open positions or how to apply for experiences such as graduate school, postdoc, or faculty positions.



To help those students interested in securing a summer internship, as well as those about to enter the mathematical sciences job market, the MAA Professional Development Committee sponsored a workshop at the Joint Mathematics Meetings 2015 to share tips on how to interview and where to find information about various job opportunities. This was followed by three speed interview sessions where attendees practiced what they had learned.

The organizers received enthusiastic responses from students and faculty alike, as well as requests to help departments re-create the workshop on their own campus. This article is an attempt to do just that.

Opportunities for Math Majors

If you are a math major, it may not be apparent where the job opportunities are. The good news is that there are a variety of resources out there to help. Check out the links at the end of this article for career information from the MAA, AMS, SIAM, and institutions such as Brigham Young University and Worcester Polytechnic Institute.

Interviewing Basics

Landing an interview means you have homework to do: preparation is key! Research the company with which you will interview. Who are they? What do they do? What can you find out about the position for which you are interviewing? Make sure you know *you.* Know the information that is on your résumé, and be able to articulate why you are qualified for the job and why you are interested in it.

Many recruiters for nonacademic positions mention that they hire math students because of their problem-solving skills, their abstract thinking abilities, and their attention to detail. In preparation for an interview, think of specific examples in which you have demonstrated these attributes. It also helps to familiarize yourself with common interview questions (see the links at the end of this article) and to think of two to three questions that you would like to ask during the interview.



In addition to preparation, how you present yourself matters. Dress for success. That means you need to look clean, neat, and professional. But it doesn't stop there. Be sure to bring a few extra copies of your résumé to the interview, as well as business cards (don't have any? get some made), a pen, and a professional folder with a notepad.

Project confident body language. Speak clearly, confidently, and professionally. Listen carefully to questions during the interview, asking for clarification if necessary. And don't forget to smile and say thank you at the end. Afterward, it would be appropriate to send a thank-you email message, as well.

Sound like a lot of advice? It is, which is why the last interview tip is: practice. Practice shaking hands and introducing yourself (your name, major, institution, year in college). Practice answering common interview questions and asking questions. Practice out loud, preferably with another person. It is also important to practice answering some questions that you haven't thought about, so have your practice partner grill you.

Do It on Your Campus

You can replicate this event for the undergraduate and graduate students at your own institution. Put together a PowerPoint summarizing some of the best practices for interviewing (above) and invite two or three employers, who hire your graduates, to share their own tips. You can choose to focus on internships, jobs in industry, graduate school, or postdoc and faculty positions, depending on the needs and interests of your

students. Recruit faculty members to serve as "employers" (if you are short on faculty, have students swap roles—serving as an interviewer in one round and an interviewee in the next) for about three 10-minute interviewing sessions. This works best if you provide participants with information on their role (for example, an employer with MathCorp interviewing for a new position in the Data Analytics Section) and about three questions each for the employer and student (see links at the end of this article for examples). You can change up the scenario and questions for each round to provide students with the opportunity to think about how to tailor their résumé for different types of opportunities. At the end of each session, be sure to give the "employers" three to five minutes to provide the students with feedback (keep it positive!) about what they did well and what they could do to improve even more in the next round. Add some food to foster a post-workshop discussion and provide a handout that lists the links, below, to online resources about how to interview and job opportunities for math majors of all types.



Helping your students be prepared to succeed in their job search, whether it is for an internship or a full-time faculty position, is a plus not only for them, but also for your department. Strong placement of your graduates is a good recruiting tool and helps you grow future supporters of your programs. And that is a win-win for everyone.

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Helpful Links

Interviewing Basics

Examples of good job interview questions and answers: *http://abt.cm/1aWPVqv* and *http://bit.ly/1pPBbaM*

Tips on preparing for a job interview: CBS, 9 Tips: *http://bit.ly/1pPBbaM*

Advice on answering questions during a job interview: Fast Company *http://bit.ly/1eTBAyc*

Information on effective networking strategies: *http://bit.ly/1WvRaHI*

Information on Careers in the Mathematical Sciences

"Careers," MAA, maa.org/careers

"Mathematical Sciences Career Information," AMS, *ams.org/careers/*

"Careers in Math," SIAM, *siam.org/careers/thinking/pdf/brochure.pdf*

"Thinking of a Career in Applied Mathematics?" SIAM, *siam.org/careers/thinking.php*

"Choosing Math Just Adds Up," Brigham Young University, *http://bit.ly/1VY2Nby*

"Career Outlook," Worcester Polytechnic Institute, *http://bit.ly/1r3cF6R*

Information on Interviewing for Postdoc or Faculty Positions

PowerPoint presentations from Louisiana Tech University's Graduate Student Seminar Series on the elements of an application package (cover letter, résumé, teaching statement, research statement), as well as how to hunt for positions and how to conduct an effective interview: *http://coes.latech.edu/owise/owise_seminar.php*

Materials from Purdue's Online Writing Lab (OWL) on job search writing, including the application, application letters, and résumés: *owl.english.purdue.edu/owl/section/6/*

The University of Texas at El Paso's Academic Job Search Materials, including letters of recommendation, sample materials, and a resource list: *http://sa.utep.edu/careers/career-planning-resources*

Job Search Materials for Graduate Students and Postdocs at the University of Pennsylvania's Career Services website, including sample résumés, cover letters, PowerPoint presentations, and guides: *vpul.upenn.edu/ careerservices/*